

Original synthetic report

Patient name: Lena Varnell
Medical record number: 48271639
Date of birth: 06/18/1965
Ordering provider: Mira Norvale, MD
SURGICAL PATHOLOGY REPORT
Accession: SP32-041728
Collected: 09/10/2032 09:25
Received: 09/10/2032 10:12
Reported: 09/12/2032 15:40

SPECIMEN
A. Right upper lobe lung, image-guided core biopsy

FINAL DIAGNOSIS
Lung, right upper lobe, core biopsy:
Pulmonary adenocarcinoma.

MICROSCOPIC DESCRIPTION
Core fragments contain infiltrating malignant glands with focal papillary architecture, nuclear enlargement, and conspicuous nucleoli. Focal necrosis is present. No definitive squamous differentiation is identified in the sampled tissue.

IMMUNOHISTOCHEMISTRY
TTF-1: Positive in tumor nuclei.
Napsin A: Positive in tumor cytoplasm.
p40: Negative in tumor cells.
The findings support pulmonary adenocarcinoma.

CLINICAL HISTORY
Known EGFR-mutated lung adenocarcinoma. Repeat biopsy for molecular reassessment; correlate with prior therapy and testing.

GROSS DESCRIPTION
Four tan-white cores, 0.3-0.9 cm in length and up to 0.1 cm in diameter; submitted entirely in cassette A1.
Fixative: 10% neutral buffered formalin.

MOLECULAR TESTING
Block A1 selected; estimated viable tumor cellularity 40%.
Targeted DNA sequencing reported separately as MP32-006381.
The diagnosis is based on this biopsy; pathologic stage is not assigned.

Electronically signed: Arlen Voss, MD
Lena Varnell | 48271639 | SP32-041728 | Page 1 of 1

Institution-developed pipeline

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**** | ****-***** | Page 1 of 1

Refined local rules

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Accession: *****

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MOLECULAR PATHOLOGY REPORT
Accession: MP32-006381
Parent accession: SP32-041728
Collected: 09/10/2032 09:25
Received: 09/13/2032 11:05
Reported: 09/18/2032 14:20

SPECIMEN INFORMATION
Site: Right upper lobe lung
Diagnosis: Pulmonary adenocarcinoma
Material: Formalin-fixed paraffin-embedded core biopsy
Specimen part: A | Block: A1
Tumor cellularity: 40% | Macrodissection: Performed

MOLECULAR RESULTS
EGFR p.L858R detected; variant allele fraction 28%.
EGFR p.T790M detected; variant allele fraction 12%.

VARIANT DETAILS				
Gene	Transcript	Exon	Coding change	Protein change
EGFR	NM_005228.5	21	c.2573T>G	p.L858R
EGFR	NM_005228.5	20	c.2369C>T	p.T790M

INTERPRETATION
Two EGFR substitutions are detected. Correlate with prior molecular testing and treatment history. The phase relationship of the two variants is not determined. Tumor-only testing does not establish whether a variant is somatic or germline.

QUALITY METRICS
DNA input: 42 ng | Mean target depth: 1540x
L858R locus depth: 1640x | T790M locus depth: 1240x
Target bases at >=250x coverage: 99.4%
All listed quality metrics are simulated for this illustration.

METHOD
Illustrative targeted DNA sequencing of EGFR exons 18-21. Single-nucleotide variants and small insertions/deletions assessed. Illustrative reporting threshold: 5% variant allele fraction. No RNA fusion or copy-number assessment is included.

LIMITATIONS
Low tumor content, sampling, and variants below the reporting threshold may limit detection. A negative result would not exclude alterations outside the interrogated regions. No paired normal sample was analyzed. Assay specifications are illustrative only.

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Lena Varnell | 48271639 | MP32-006381 | Page 1 of 1

Institution-developed pipeline

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MOLECULAR RESULT - EXACT EXCERPTS

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EGFR p.T790M detected

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EGFR p.***** detected

Refined local rules

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